

Lowest power BLE solution(w/energy harvesting)

December 2022

Atmosic Organization

- Experienced Wireless Systems-on-a-Chip (SoC) team of 130+
- Strong wireless SoC technologies
- Strong focus on end to end systems engineering
  → optimizing solutions for compelling use cases
- Established track record of shipping billions of chips
  → focus on customer enablement
- Engineering Locations: Campbell, CA + Irvine, CA + Taiwan + China + Paris



David Su CEO

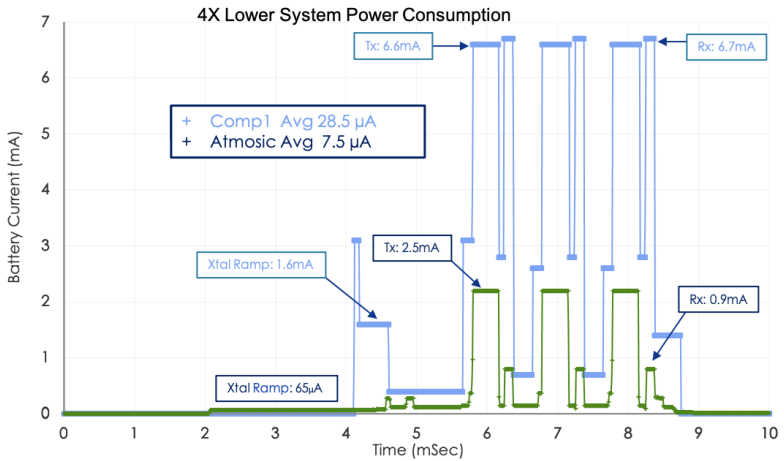
Masoud Zargari COO	Srenik Mehta Eng	Mike Fortin Sales	Srinivas Pattamatta Marketing
• RF • SoC • Operations • Finance	• Architecture • Hardware Systems • Software • App Eng • TW Eng	• North America • Asia • China • EU • Field App Eng	• Marcom • Prod Marketing

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Differentiation points from others

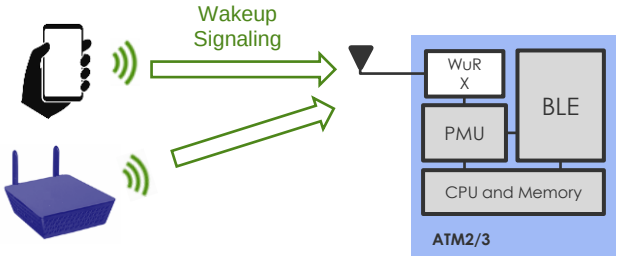
Lowest Power

5-10X Lower than competitor



On-Demand Wakeup

Enable Hibernation wake up over RF
Input Freq 400Mhz ~ 2.5Ghz



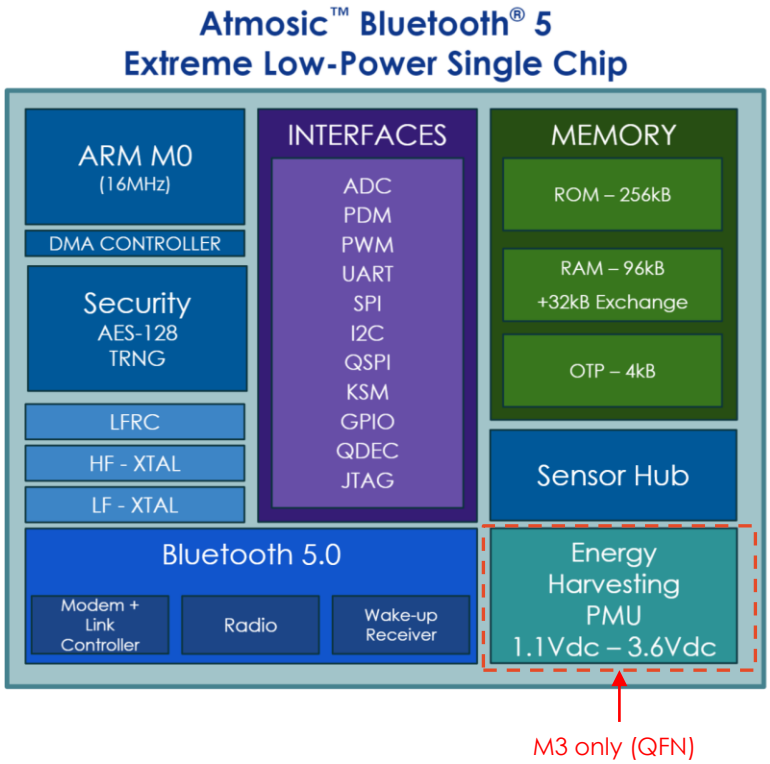
Managed Harvesting

PV or RF Harvesting to enable Forever
Battery Life or Battery Free



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Gen 1(ATM2/3 Series) – In production



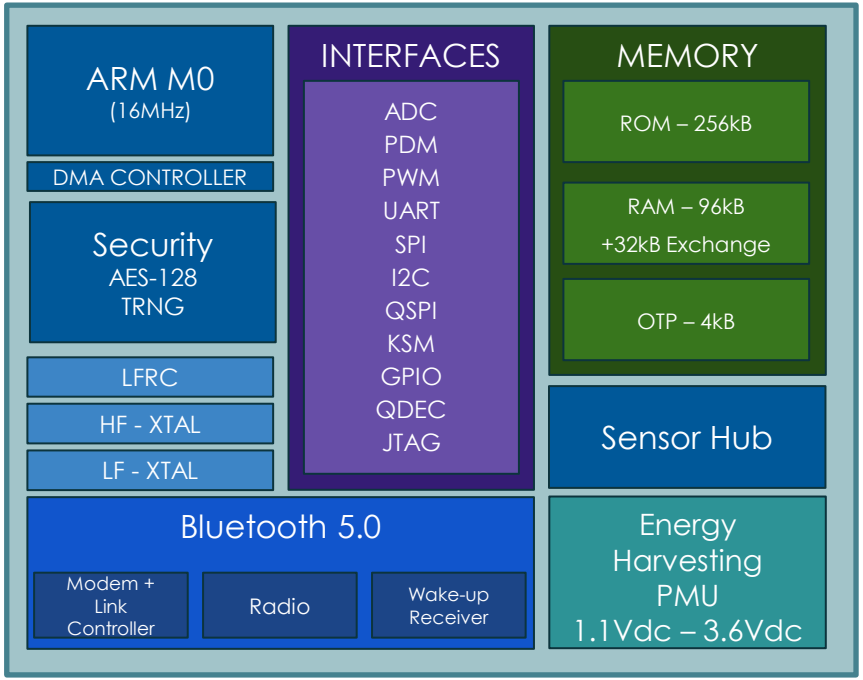


- **Lowest Power Bluetooth 5.0 SoC**
  - Active Rx (-94dBm) : **0.9mA** @3V
  - Active Tx (0dBm) : **2.5mA** @3V
- **On-Demand Wakeup**
- **Managed Energy Harvesting**
- **Sensor Hub: Event-Based Power Optimization**
- MCU: ARM Cortex® M0
- Memory: 128kB RAM, 256KB ROM, 4kB OTP, 1MB Flash
- Interfaces: I2C, SPI, HCI UART, ADC, PWM
- Packaging: QFN(5x5, 6x6mm), WLCSP
- **BT SIG Qualified & Shipping**

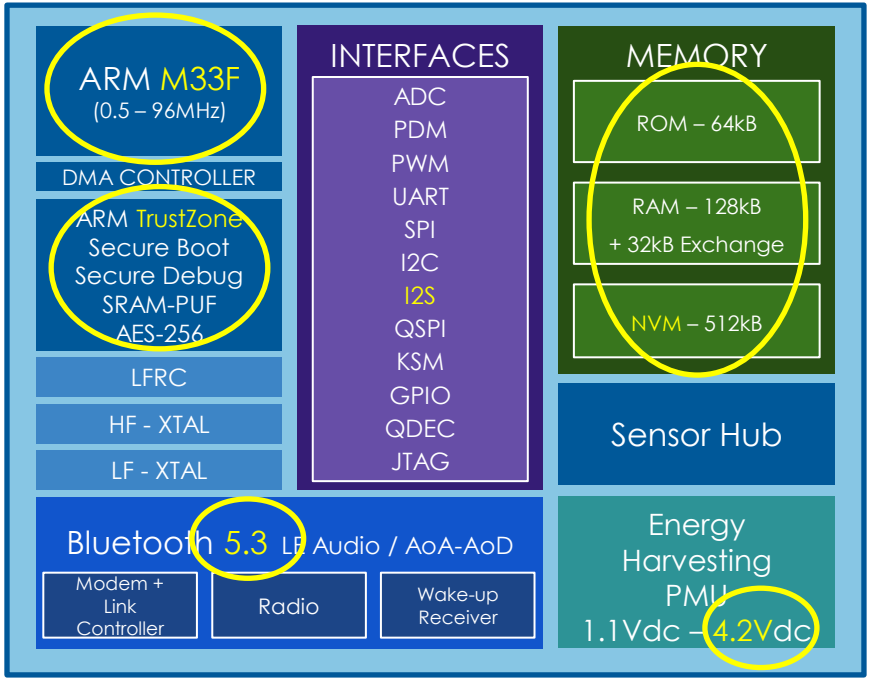
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Gen1(ATM2/3) vs Gen2 (ATM33/e)

ATM2/3

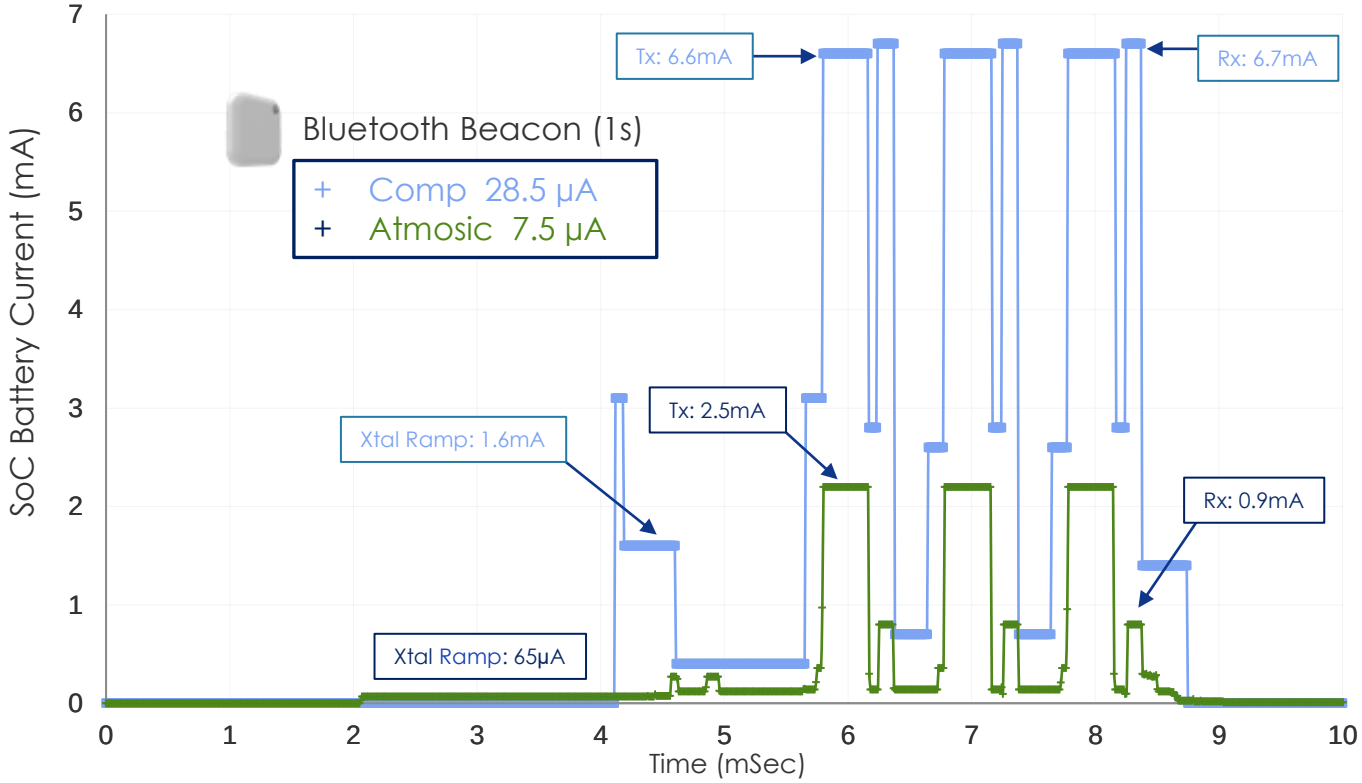


ATM33/e



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Lowest Power (4X Lower Power Consumption)



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Atmosic vs Competition

	Atmosic	Nordic (nRF52832)	Nordic (nRF52840)	Dialog (DA14531)	Dialog (DA1458x)	Dialog (DA1468x)	Cypress (CYW2073x)	Telink (TLSR8255)	Renesas (RL78/G1D)	OnSemi (RSL10)	ST Micro (BlueNRG-2)	TI (CC2642)	Microchip (IS1870/71)
System (RX) @ -93dBm @3V	0.9mA	6.7mA	4.8mA	2.2mA	5.1mA	5.8mA	9.8mA	8mA	3.5mA	3mA	7.7mA	6.8mA	12mA
System (TX) @ 0dBm @3V	2.5mA	6mA	6.4mA	3.5mA	4.8mA	5.1mA	9.1mA	6.7mA	4.3mA	4.6mA	8.5mA	7.5mA	12mA
Application Energy Saving	Yes	NOT SUPPORTED											
MCU Power Optimization	Integrated												
Wake on Receive	Integrated												
Energy Harvesting	Integrated	No	No	No	Ext Chip	Ext Chip	No	No	Ext Chip	No	No	No	No

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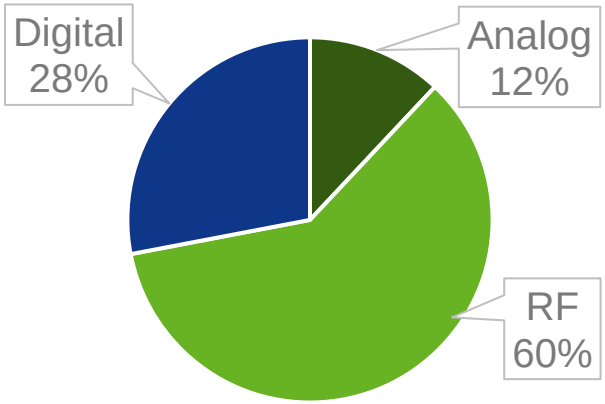
Lowest Power Consumption Protected by Patents

Competitors → **Atmosic™**

Applicable to any Wireless Standard

Digital

- Adaptive scaling of supply voltage (Sub-threshold)
- Applicable to Digital only ~1/3 of the power consumption



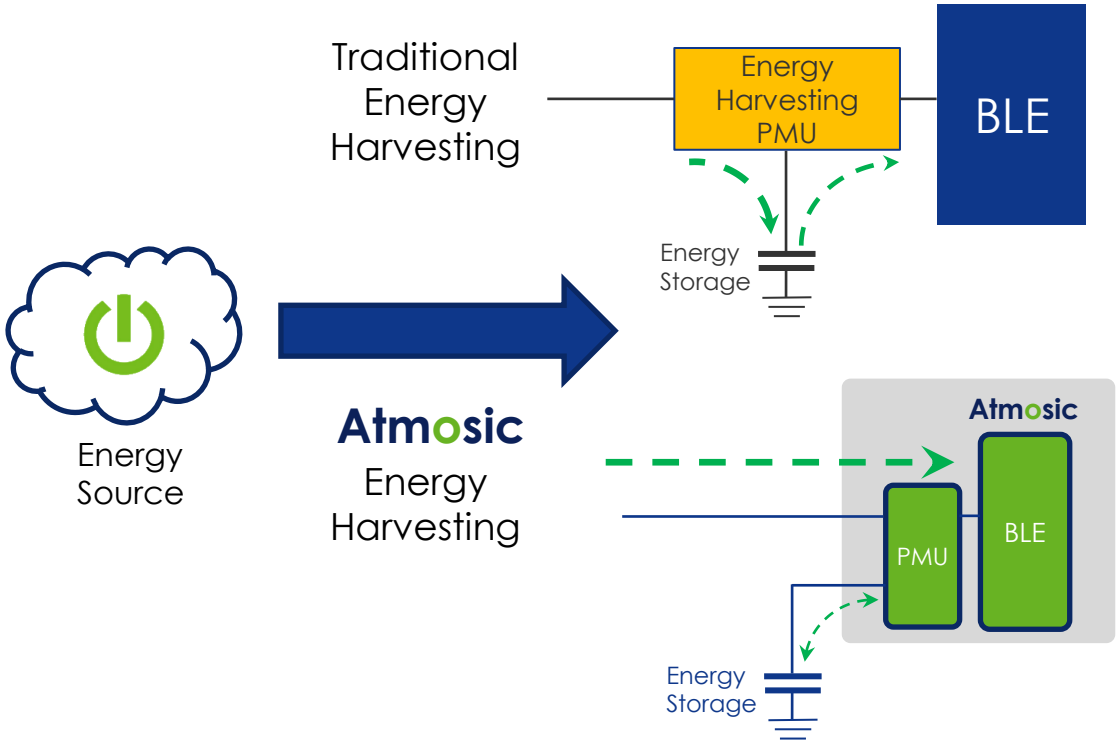
Power Consumption of typical Radio SoC

RF & Analog + Digital

- Multi-dimensional adaptive scaling
- Demonstrated >4X Longer Battery Life

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The Atmosic Energy Harvesting Difference



- Harvested energy must be stored prior to use
- Higher BOM, Lower overall efficiency
- Higher power consumption requires larger harvester

- Direct use of harvested energy
- Auto recovery from brownout
- Supports multiple energy sources simultaneously

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Revolutionizing IoT Smart Devices



Keyboard



ESL



Beacon



IoT Module





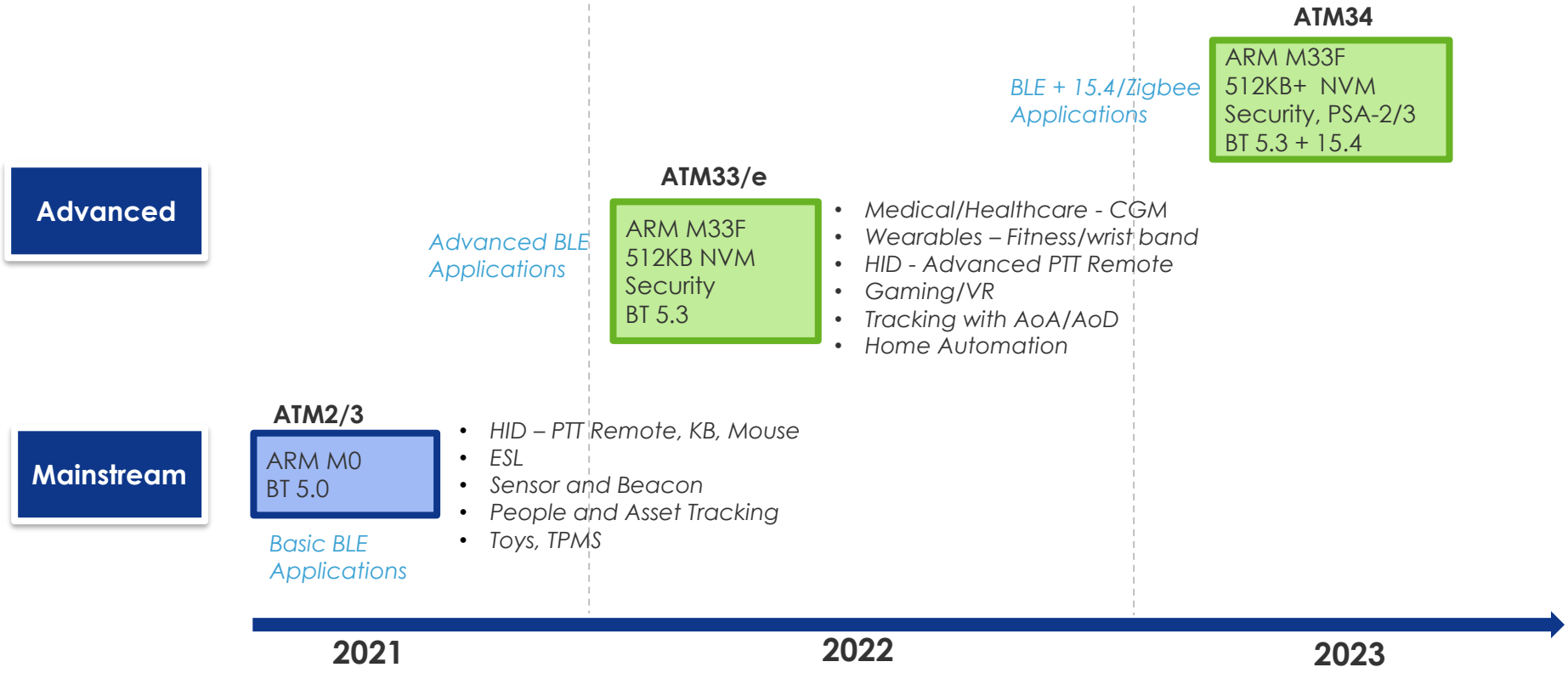
Contact Tracing



Push-to-Talk
Remote Control

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Product Roadmap



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Thank you!

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